



SECJ3553 ARTIFICIAL INTELLIGENCE

Section 03

ASSIGNMENT 4 PROOF OF CONCEPT (10%)

Theme: Agriculture

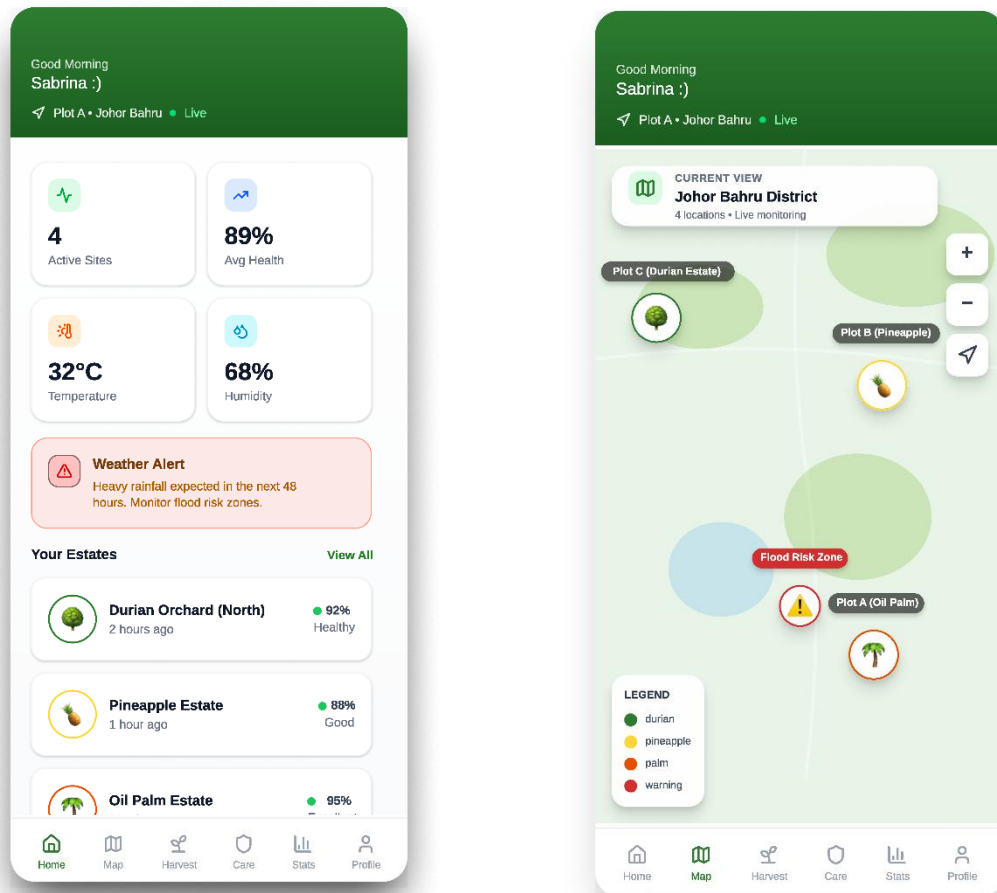
AI-Powered Predictive Analytics for Optimised Crop Yield and Risk Mitigation

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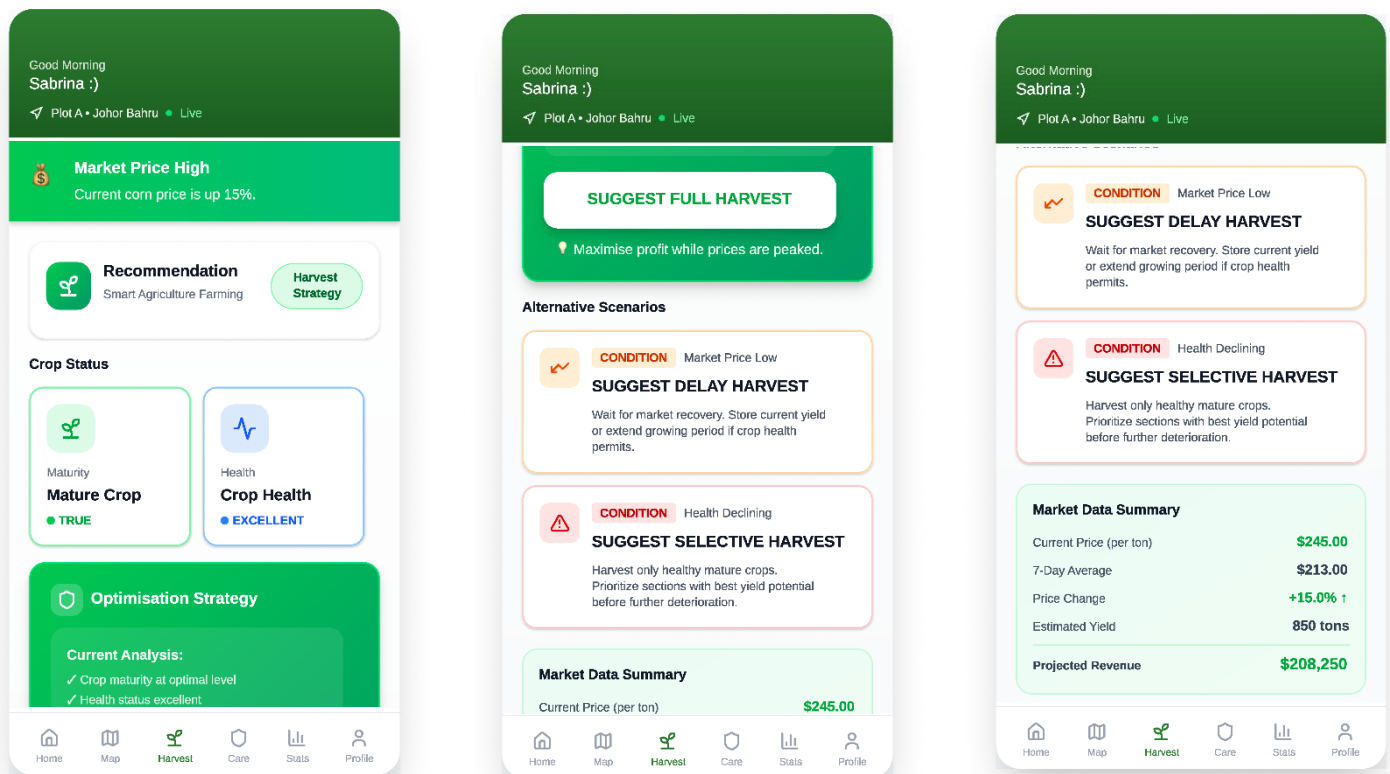
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SCENARIO 1 Critical Flood Risk Management



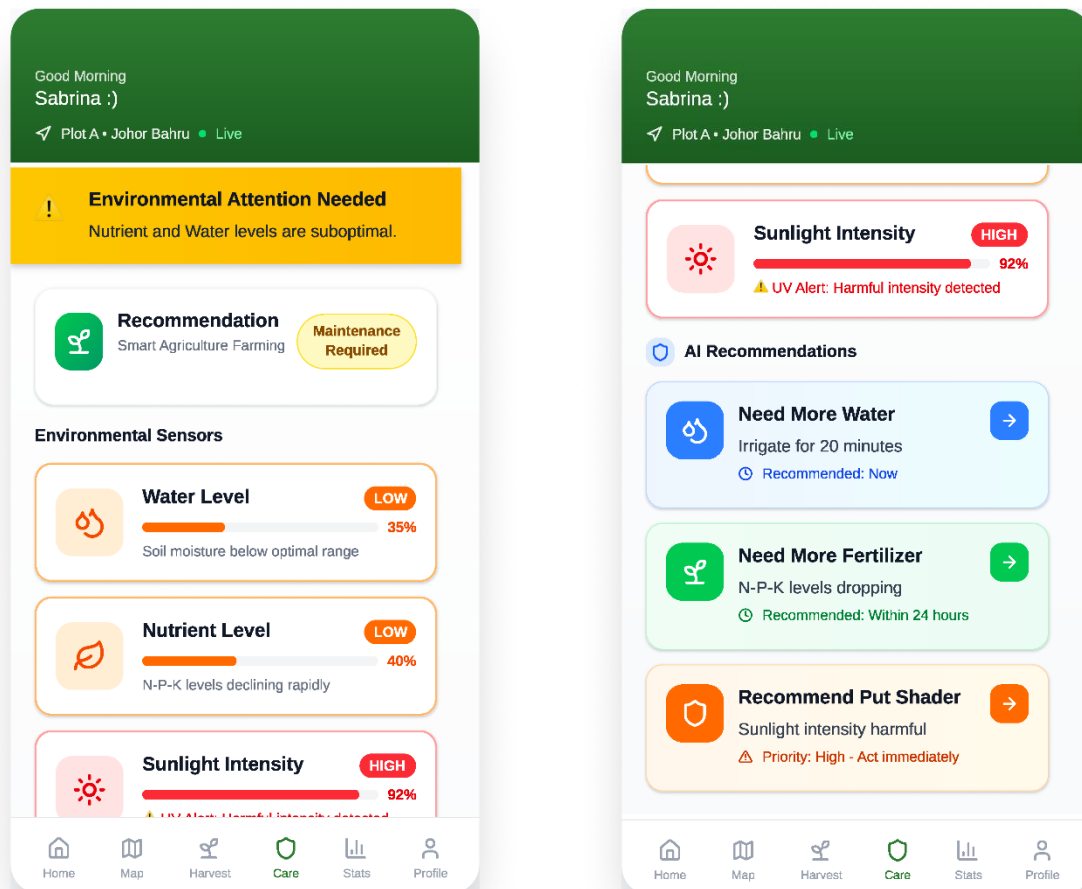
Performance (P)	The primary goal is minimising loss. It is measured by the agent's ability to warn the farmer before flood occurs.
Environment (E)	The agent detected heavy rainfall expected state for the next 48 hours. It will mark it as a Flood Risk Zone on plot A on the map.
Actuator (A)	It will show a red weather alert on the dashboard. The specific danger zone is highlighted red on the map.
Sensors (S)	The system will obtain data from the weather API for predictions and soil sensors for soil moisture.

SCENARIO 2 Harvest Optimisation Strategy



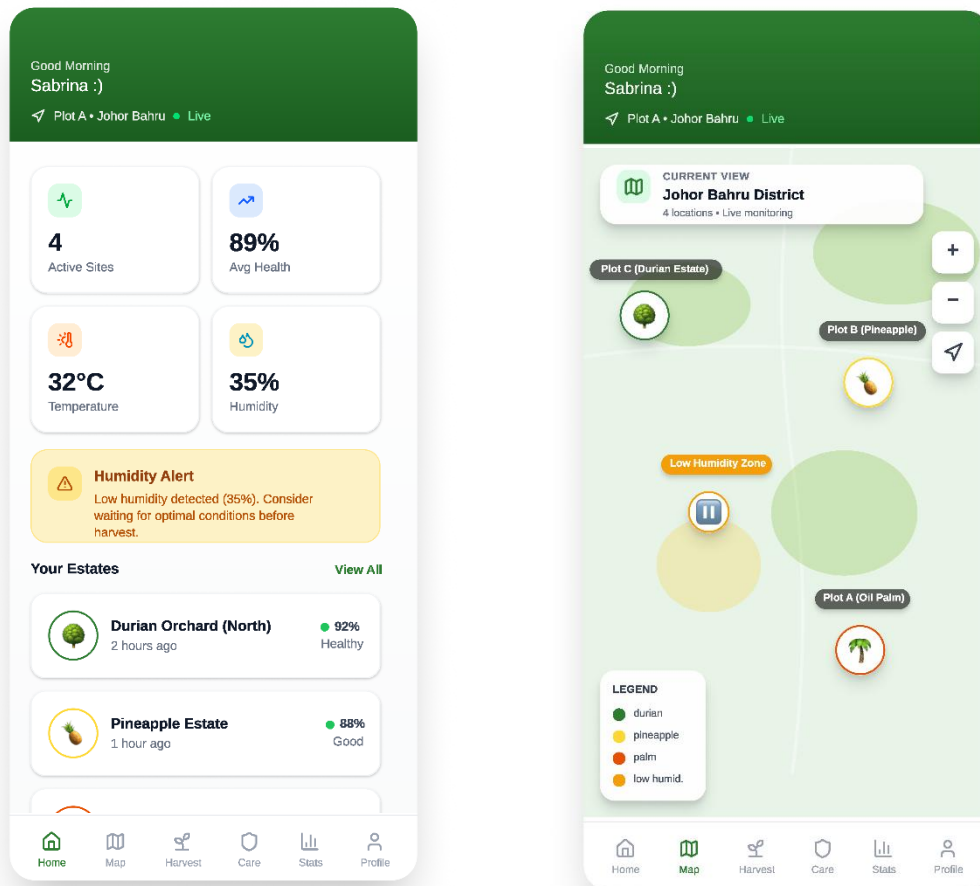
Performance (P)	The primary goal is maximising profit. The agent will trigger the farmer to harvest when the market price is at the peak.
Environment (E)	The market corn price is high. Status of the crop is mature, and the health is excellent.
Actuator (A)	It will recommend the farmer by displaying the notification Suggest Full Harvest. It will display text justification below to notify the farmer the best time to harvest.
Sensors (S)	Market data for continuous track prices. Crop monitoring to track the growth stage and crops health.

SCENARIO 3 Environmental Care & Recommendation



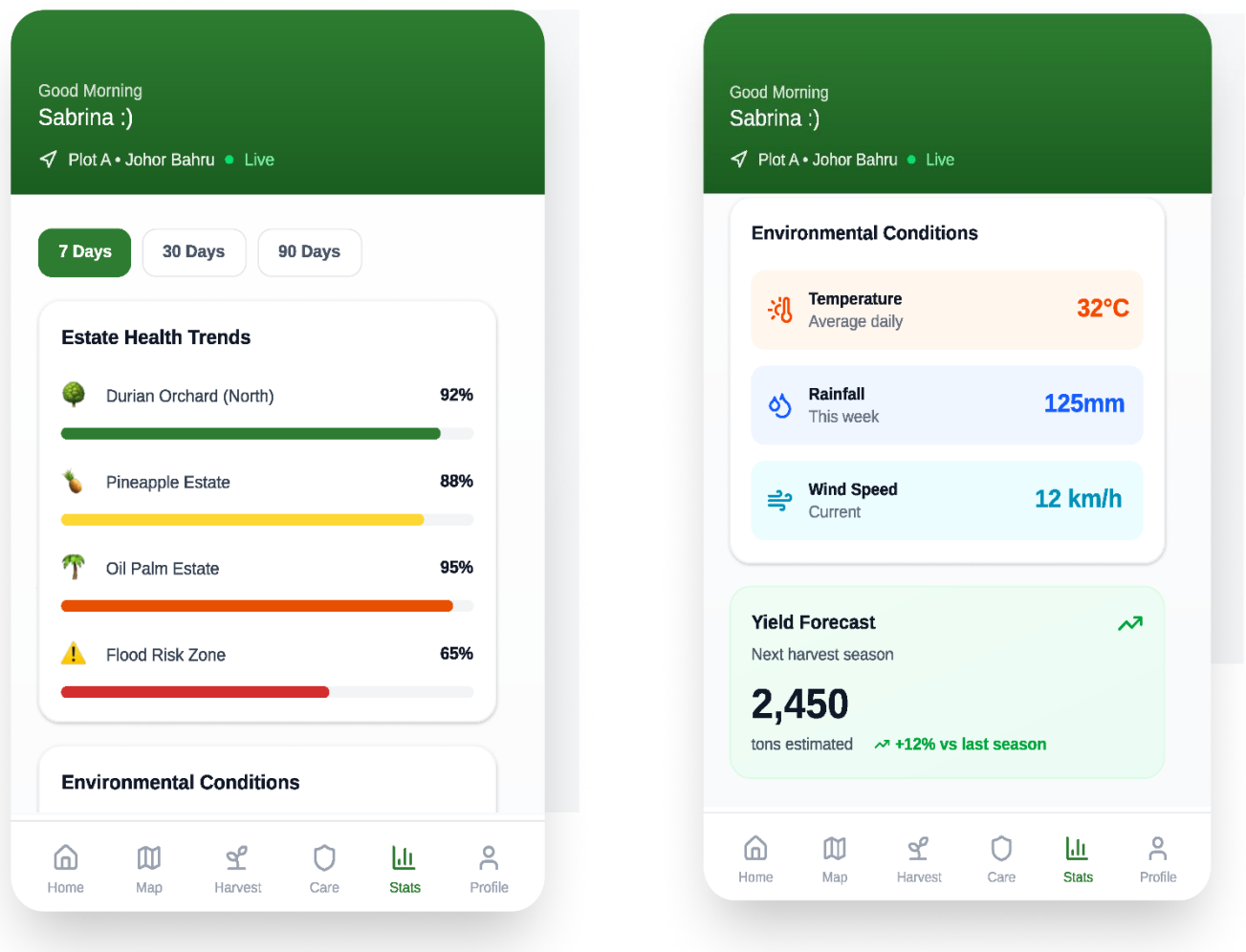
Performance (P)	The primary goal is stabilising the crop health. The agent will aim to return back the environmental variables to the optimal range.
Environment (E)	The agent detects the abnormal readings in the environment. For example, in this scenario, low water level, low nutrient level and high sunlight intensity.
Actuator (A)	The system will generate AI recommendation and a task list for the farmer. For example, need more water and recommend put shader.
Sensors (S)	Soil moisture sensor to detect the water content. Nutrient sensor to monitor soil chemical content and light sensor to measure the sunlight intensity.

SCENARIO 4 Low Humidity, Wait for Action



Performance (P)	The primary goal is making sure the quality is high. If the farmer harvests the crop in low humidity, it will lower the quality of the crops.
Environment (E)	The agent detected a specific area is having a low humidity zone.
Actuator (A)	It will show an alert. For example, in this scenario, it suggests consider waiting for optimal condition.
Sensors (S)	Humidity sensors are used in the plots to detect the humidity of the crops that dropped below the standard humidity level.

SCENARIO 5 Performance Analysis & Yield Prediction



Performance (P)	The primary goal is prediction accuracy and yield improvement. The system will track if the actual crop yield meet the predicted forecast growth.
Environment (E)	The temperature trend, rainfall logs and wind speed over the 90 days.
Actuator (A)	The system will show a visual graph that show the health trend and yield forecast. It will also display the estimated yield forecast to help the farmer plan.
Sensors (S)	The system database to aggregate the past data from all sensors to compute the trends and future predictions.